Paper A-02: Earth Atmospheric Circulation: A Constraint-Driven Flux Dynamics (CDFD) Perspective

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May 2026

Abstract

This paper treats atmospheric circulation as a CDFD/AFL modelling hypothesis. Solar heating, latent heat, water vapour, aerosols, and momentum are read as driving fluxes Φ moving through pressure gradients, rotation, stratified layers, topography, and boundary-layer drag. The purpose is not to replace meteorology or climate dynamics. The purpose is to name the measurable flux, constraint, responsiveness, and memory terms that make atmospheric overload or stagnation testable.

Symbol Conventions

CDFD/AFL Part IV symbol convention.

Symbol	Part IV meaning	Cross-domain reading
Φ	Driving flux or throughput	Energy, matter, signal, capital, information, traffic, force, or biological load entering a system
C	Active constraint or capacity burden	Bottleneck, resistance, boundary, scarcity, toxicity, regulation, topology, latency, or load-bearing limit
S	Responsiveness or adaptive surface	The system's ability to reconfigure, buffer, route, repair, learn, or preserve function under changing load
M_s	Structural memory	Stored damage, hysteresis, inherited topology, institutional memory, trained weights, ecological legacy, or retained state
Ψ_s	Adaptive operating ratio $(\Phi/C)\cdot S\cdot M_s$	Local bookkeeping gauge for constrained operation, overload, recovery, or regime change; not a universal constant
Λ	Life Number or capstone surplus ratio where explicitly defined	Reserved for Part II-derived life-number notation or synthesis-level surplus measures, never an undefined decoration

Greek-symbol convention.

Symbol	Local role	Interpretation rule
α	Accumulation or reinforcement coefficient	Sets how fast a pathway, constraint, habit, cascade, or retained structure grows under load
β	Relaxation, decay, clearance, or washout coefficient	Sets loss, recovery, damping, forgetting, degradation, or return toward baseline
γ	Coupling, diffusion, redistribution, or transfer coefficient	Sets spread across nodes, layers, tissues, markets, infrastructures, media, or physical phases
δ, ϵ	Perturbation, tolerance, small offset, or numerical guard	Used only as local thresholds, stability margins, measurement tolerances, or stress increments
η, λ, μ, ν	Efficiency, rate, baseline, intensity, or frequency terms	Meaning is fixed by the equation, subscript, and domain-specific proxy in the paragraph where the term appears
ρ, σ, τ, ϕ	Density/correlation, variability/noise, time-scale, phase, or field terms	Local coefficients only; subscripts bind them to the named pathway, domain, or experiment
Ω, Λ	Domain/state space; Life Number where defined	Ω names a modeled state space; Λ carries life-number or synthesis notation only when the paper defines it

1 Series Context

Part I sets the flow–constraint language used here [3]. Part II develops the Life Number and the origins-of-life use of the same notation [4]. Part III applies AFL to biology and medicine [5]. Part IV [6], DOI https://doi.org/10.5281/zenodo.20395901, asks whether the same audit language clarifies cross-domain measurement without turning analogy into evidence.

2 Atmospheric Transport Frame

The atmosphere is a moving constraint network. Uneven heating creates pressure gradients; rotation bends flow; mountains, land–sea contrast, cloud formation, surface roughness, and radiative forcing reshape the pathways available to air and moisture. CDFD/AFL treats those pathways as an audit surface: where does flux rise, where does the atmosphere lose room to adjust, and where does memory from prior heat, moisture, or aerosol loading still shape the next state?

3 Candidate Variable Map

CDFD term	Atmospheric reading	Example proxy
Φ	heat, moisture, aerosol, or momentum flux	radiative flux, latent heat, wind stress
C	active circulation constraint	inversion strength, topography, drag, stability
S	responsiveness	convective adjustment, mixing, storm-track mobility
M_s	structural memory	soil moisture, sea-surface heat, aerosol persistence
Ψ_s	operating ratio	ventilated, stagnant, amplified, or overloaded regime

4 Overload and Stagnation

An atmospheric overload claim is meaningful only when the flux and constraint are named together. A heat-wave example might use radiative and sensible heat flux as Φ, boundary-layer stability and weak advection as C, convective mixing as S, and stored soil or ocean heat as M_s. A pollution-stagnation example might use emissions as Φ, inversion strength as C, turbulent mixing as S, and accumulated aerosol load as M_s. These mappings remain hypotheses until compared with atmospheric observations and standard models.

5 Measurement Layer

The measurable layer comes first. A useful atmospheric application gives Φ, C, S, and M_s observable proxies, a spatial grid or network, a time window, and a failure condition such as persistent stagnation, blocked moisture transport, convective inhibition, or extreme heat amplification. If conventional fluid-dynamical variables explain the event without added value from CDFD/AFL, the mapping adds no release-level claim.

6 Current Runtime Diagnostic

The Part E diagnostic run includes a climate/atmosphere adapter row and the universal cascade stress test. These files check finite runtime behavior and regime reporting in the current CDFD Runtime [7]. They are a model audit for later measurement, not a replacement for atmospheric data.

7 Tests That Can Break the Mapping

The atmospheric mapping weakens if measured flux and constraint proxies do not track stagnation, amplification, or recovery; if the same event is explained more cleanly by established atmospheric diagnostics; or if the CDFD/AFL terms cannot be separated from ordinary variable names in a reproducible analysis.

8 Major Universal Upgrade: Mechanism, Scale, and Tests

8.1 Observable Translation

The paper-specific translation begins with an observable chain rather than a verbal analogy. For Earth Atmospheric Circulation, the proposed drive is differential solar heating, latent heat, and moisture transport. The active limitation is static stability, friction, topography, and radiative limits. Responsiveness is carried by convection, waves, circulation, and cloud adjustment, while retained state is represented by soil moisture, sea-surface temperature, aerosol, and ice states. The outcome to explain is circulation strength, precipitation, and extreme-event persistence.

Table 1: Operational CDFD/AFL map for Paper A-02.

Term	**Paper-specific observable meaning**	**Measurement question**
Φ	differential solar heating, latent heat, and moisture transport	What rate, load, gradient, or demand enters the focal system?
C	static stability, friction, topography, and radiative limits	Which measured bottleneck limits transfer, function, or recovery?
S	convection, waves, circulation, and cloud adjustment	Which process changes effective capacity or reroutes the load?
M_s	soil moisture, sea-surface temperature, aerosol, and ice states	Which retained state makes equal present loads produce different futures?
Y	circulation strength, precipitation, and extreme-event persistence	Which outcome is predicted out of sample, with uncertainty?

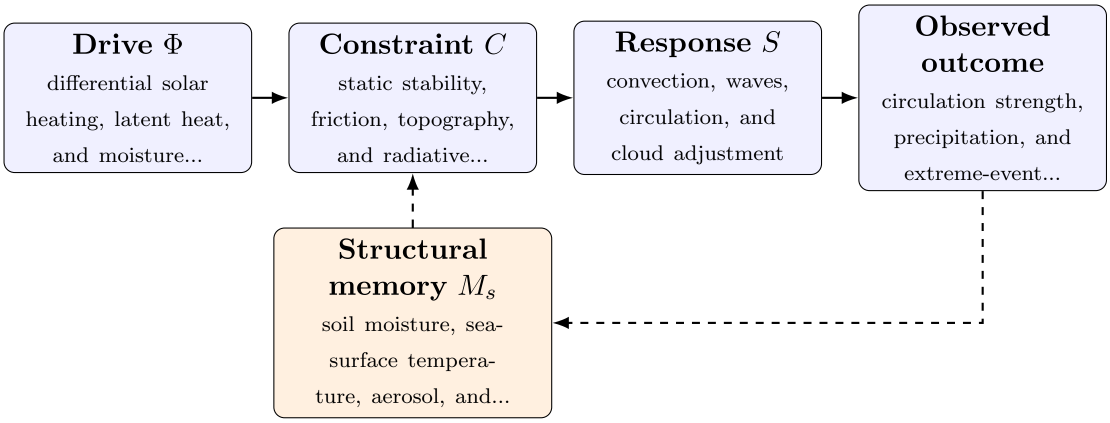

Figure 1: Paper A-02 observable CDFD/AFL architecture for Earth Atmospheric Circulation. Solid arrows give the proposed forward chain; dashed arrows make history-dependent feedback explicit.

8.2 Minimal Coupled Model

A paper-level model must separate throughput, constraint accumulation, response, and history. One deliberately minimal form is

$$Y(t) = \frac{\Phi(t)S(t)}{\epsilon + C(t)}, \tag{1}$$

$$\frac{dC}{dt} = \alpha\Phi(t) - \beta S(t)C(t) + \gamma\mathcal{L}C(t), \tag{2}$$

$$\frac{dM_s}{dt} = \eta C(t) - \mu M_s(t), \tag{3}$$

$$\Psi_s(t) = \frac{\widehat{\Phi}(t)}{\epsilon + \widehat{C}(t)}\,\widehat{S}(t)\,[1 + \omega\widehat{M_s}(t)]. \tag{4}$$

Here Y is the measured output, $\epsilon > 0$ prevents a singular denominator, $\mathcal{L}$ is a spatial or network coupling operator, and hats denote explicitly normalized variables. The sign and magnitude of ω must be estimated: memory can preserve useful organization or amplify damage. No fixed threshold is universal before the variables, normalization, and observation scale are declared.

For this paper, the hypothesized causal sequence is specific. A change in differential solar heating, latent heat, and moisture transport arrives faster than convection, waves, circulation, and cloud adjustment can compensate. This raises or redistributes static stability, friction, topography, and radiative limits. If the load persists, the retained state encoded by soil moisture, sea-surface temperature, aerosol, and ice states changes the next response, so a later event of equal magnitude can produce a different trajectory in circulation strength, precipitation, and extreme-event persistence. The scientific task is to estimate that sequence against the best ordinary model of the domain, not merely to redescribe the outcome after it occurs.

8.3 Cross-Scale Universality Without Mechanistic Erasure

For the Earth systems family, the universal comparison is between coupled transport, finite environmental capacity, buffering, and retained landscape or ecosystem state. Universality here cannot erase conservation laws, geometry, or scale; it must expose where those domain equations create comparable overload and recovery structures. [2, 8, 1] The CDFD programme is cumulative: Part I supplies the flow–constraint foundation [3]; Part II asks when maintained throughput supports persistent organized states [4]; Part III develops adaptive limitation and biological memory [5]; and Part IV [6] tests how much of that architecture survives translation. The CDFD Runtime [7] supplies a reproducible numerical stress surface, but it does not substitute for the domain evidence named here.

The required scale declaration for this paper is minutes to decades; boundary layers to planetary circulation. At short scales, the key question is whether the incoming drive is transmitted, buffered, or diverted. At intermediate scales, response changes effective capacity and network exposure. At long scales, retained structure can alter the state space itself. A result is genuinely cross-scale only when the aggregation rule is stated and the same conclusion is not an artifact of averaging.

8.4 Evidence Ladder and Discriminating Tests

Table 2: Evidence ladder for Paper A-02. Each rung can fail independently.

Test	Design	Discriminating result
Proxy audit	Measure reanalysis, radiosondes, satellite radiation, and precipitation products and preregister how each record maps to Φ, C, S, M_s, and Y.	The mapping fails if proxies are non-identifiable, circular, or change meaning across cases.
Load ramp	Apply or observe a sustained heat and moisture anomaly while estimating the dominant constraint and response time.	A threshold claim requires a reproducible nonlinear change and uncertainty bounds, not a visually chosen breakpoint.
Recovery and memory	Match present load across units with different histories, then compare recovery paths.	The memory claim requires history-dependent divergence after current conditions are controlled.
Model comparison	Compare the baseline domain model with and without the CDFD/AFL state variables using held-out data.	The paper's stated falsifier is: the mapping cannot predict circulation or recovery residuals beyond standard atmospheric equations.

The strongest practical design combines a prospective perturbation with a recovery window. The load-ramp phase estimates where constraint begins to rise faster than output. The recovery phase estimates β and μ, separating fast relaxation from persistent memory. A topology or coupling intervention then asks whether rerouting changes the cascade as predicted. Null results are informative because they identify domains in which the universal notation adds no measurable value.

8.5 Research Programme and Boundary Conditions

The immediate research programme has four steps. First, construct a documented dataset from reanalysis, radiosondes, satellite radiation, and precipitation products and retain raw units before normalization. Second, fit a baseline domain model and report its calibration and out-of-sample error. Third, add the smallest identifiable CDFD/AFL state extension and test whether it improves prediction of circulation strength, precipitation, and extreme-event persistence. Fourth, repeat the analysis across the declared scale range and publish negative cases as part of the universality audit.

Three boundaries are non-negotiable. Correlation between flow and failure does not identify a constraint mechanism. A fitted memory term does not prove that the stored state has the same material meaning in another domain. Finally, a runtime cascade is evidence about the implementation, not evidence that the real system follows it. The paper becomes stronger when it can say precisely where the translation stops.

9 Conclusion

Atmospheric circulation can be read as constrained transport when the variables are tied to observations. The CDFD/AFL contribution is a common audit language for flux, bottleneck, response, and memory. Its value depends on whether that language sharpens measurable atmospheric questions.

Conflict of Interest

The author declares no conflict of interest.

Data Availability

Part-specific runtime diagnostics are under each Part's `outputs/` directory.

Release-wide code: `Part_E_Synthesis/supplementary/`.

Generated summaries: `Part_E_Synthesis/outputs/`.

Figures: `Part_E_Synthesis/figures/`.

10 Reference Layer

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